Ricardo Emmanuel Uriegas Ibarra

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Resume

AI/ML Engineer with one year of professional experience designing and implementing scalable APIs and AI/ML systems. Proven ability to optimize performance, automate workflows and build secure software solutions using AI efficiently.

Experience

Software Developer II June 2026 - September 2026 (Caused by layoffs)
Oracle Cloud Infrastructure Team *Mexico*

- Evaluate requirements through Agile to ensure the viability of solution proposals for customers
- Design software components and LLM workflows for a full-stack platform to optimize the integration of model fine-tuning, dataset generation, reinforcement learning and inference.
- Prioritize dependencies in the development process using Agile to reduce rework and development times.
- Validate software functionality through unit and functional testing reducing the risk of undetected implementation discrepancies in the future and producing early corrections.
- Update technical documentation for software components, LLM workflows and implementation decisions to ensure technical traceability and decrease software maintenance.
- Assess feasibility of Large Tabular Models (LTMs) and their practical applications (such as fraud detection, prediction, etc), benefits and drawbacks in real scenarios to improve Oracle data product
- Contributed to reduce 75% base-model storage requirements in model fine-tuning workflows by the use of LLM models quantization
- Planned a 6.2% relative ROUGE-2 improvement over standard LoRA by integrating AdaLoRA's adaptive parameter allocation into fine-tuning workflows

Software Developer September 2025 - December 2025
Center for Research and Advanced Studies of the IPN (CINVESTAV) *Mexico*

- Design a REST API microservices on Docker with PHP and PostgreSQL for a medical platform to ensure findability, accessibility, interoperability and reusability of the data products
- Develop a distributed search engine across databases with multi-criteria filtering, input sanitization, rate limiting, OWASP HTTP headers, pagination and metadata to optimize catalog retrieval and metadata reuse
- Manage containerized service deployment with Docker Compose, health checks and persistent logging to ensure availability and traceability of microservices

AI Engineer September 2024
Center for Research and Advanced Studies of the IPN (CINVESTAV) *Mexico*

- Design a convolutional neural network (CNN) architecture with dropout regularization for audio spectrogram classification to optimize speech and music discrimination for a database full of audios
- Develop TensorFlow/Keras training pipelines with image resizing and feature normalization to ensure consistent model inputs across a 10,310-sample audio dataset
- Validate model generalization through accuracy and loss monitoring with early stopping to avoid overfitting
- Optimized speech-music classification accuracy to 84% on 2,868 held-out audio spectrograms by implementing a CNN with dropout regularization and input normalization

Research Intern September 2023
National Institute of Electricity and Clean Energy (INEEL) *Mexico*

- Develop R scripts for causal structure learning with the PC, FCI and RFCI algorithms (pcalg) to ensure reproducible generation of causal graphs (CPDAGs) from tabular data
- Design data preprocessing pipelines (categorical encoding and Gaussian transformation) to mitigate violations of conditional independence test assumptions
- Eliminated memory constraints in PC algorithm execution over 49,019 records by replacing the discrete independence test with a Gaussian test through bestNormalize transformations

Math Teacher Assistant September 2021 - June 2022
Center for Technological, Industrial and Service Studies No. 271 *Mexico*

Education

B.S. Information Technology Engineering September 2022 - December 2025
Universidad Politécnica de Victoria
Programming Technician July 2019 - August 2022
CBTIS 271 *Mexico*

Papers

Free Alternative for Opinion Mining on Twitter: Integration of Nitter, Puppeteer and VADER December 2025
Published here *DGUTyP*
Developed a free and legal sentiment analysis tool for posts on the social network X, without relying on its official API. The proposed tool uses web scraping via Nitter to collect content, filter by language, and perform sentiment classification using the VADER model. Results are visualized through charts and automated clustering. In tests with the term "Liquid Glass," the system demonstrated acceptable accuracy, achieving an overall F1-score above 81%. This approach offers a cost-free, replicable alternative for researchers or organizations facing economic barriers to accessing social data.

Academic Projects

SQL Interpreter, IDE and Modeler from Scratch
Interpreters *Java Vanilla, Maven*
- Oriented to csv files management
- Proficient in modeling, generating, and validating SQL scripts for execution across multiple CSV based data repositories functioning as databases.

Stone Detection for Beans
Image Processing *Python, OpenCV, PyQt6*
- Built an algorithm that can detect stones on a photo of beans
- Implemented entirely through traditional image processing, avoiding the need of heavy computational tools

Rotating Tree Fruit Detector Model
Machine Learning *Python, PyQt6, YOLO*
- Live fruit count detector model for ripe berries
- Trained with a short dataset and an accuracy of 87%

Certifications

- Google UX Design 2025
- CCNAv7: Enterprise Networking, Security, and Automation 2024
- Google Project Management 2023
- More if you click here

Languages

English Advance

Spanish Native